

Prompt 1 — Controller App

Prompt 1 — Controller App (Kotlin/Android)Đ

Smart School Bell — Live Announcement WebRTC UpgradeĐ

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****Use this prompt in Cursor inside your Controller App repository.****Đ

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TaskĐ

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Upgrade Live Announcement to WebRTC (Controller / Sender App).Đ

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ContextĐ

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This is the ****Controller App**** for Smart School Bell System. The backend has been upgraded from WebSocket PCM audio relay to ****WebRTC peer-to-peer**** live announcement.Đ

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- ****Backend base URL:**** `https://api.shikkhasomoy.com` (or local dev URL)Đ

- ****Auth:**** existing JWT `Authorization: Bearer <token>` from loginĐ

- ****Swagger:**** `https://api.shikkhasomoy.com/api-docs` !' Live tagĐ

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****IMPORTANT:**** Live audio must NOT go through the server. Backend only does ****signaling****. Audio goes directly Controller !' Player via WebRTC P2P.Đ

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What to REMOVE (old system)Đ

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Find and remove all old live announcement code that:Đ

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- Connects WebSocket `ws://.../ws/announce?role=controller` to send ****binary PCM/audio chunks****Đ

- Sends mic audio bytes over WebSocketĐ

- Uses binary message relay for live micĐ

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****Do NOT remove or break:****Đ

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- Bell schedules CRUDĐ

- Azan times CRUDĐ

- Recorded/scheduled announcements (REST `/api/announcements/\*`)Đ

- Login/authĐ

- Any other existing featuresĐ

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What to ADDĐ

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1. Dependencies (build.gradle)Đ

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```gradleÐ
implementation 'io.getstream:stream-webrtc-android:1.1.3'Ð
// OR: org.webrtc:google-webrtc (stable version for your minSdk)Ð
```Ð
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**Permissions (AndroidManifest):**Ð
Ð
- `RECORD_AUDIO`Ð
- `INTERNET`Ð
- `ACCESS_NETWORK_STATE`Ð
- `MODIFY_AUDIO_SETTINGS` (optional)Ð
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### 2. Retrofit API — LiveApiService.ktÐ
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| Method | Path | Body |Ð
|-----|-----|-----|Ð
| GET | `/api/live/config` | — |Ð
| POST | `/api/live/session` | `{ "role": "controller", "device_id": "..." }` |Ð
| GET | `/api/live/session?role=controller&session_id={id}` | — |Ð
| POST | `/api/live/offer` | `{ "session_id", "role": "controller", "offer": { "sdp", "type": "offer" } }` |Ð
| POST | `/api/live/ice` | `{ "session_id", "role": "controller", "candidate", "sdpMid", "sdpMLineIndex" }` |Ð
| GET | `/api/live/ice?session_id={id}&role=controller` | — |Ð
| POST | `/api/live/end` | `{ "session_id", "reason": "ended" }` |Ð
Ð
**POST /api/live/session response:**Ð
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```jsonÐ
{Ð
 "session_id": "uuid",Ð
 "status": "waiting_player",Ð
 "ice_servers": [{ "urls": "stun:stun.l.google.com:19302" }]Ð
}Ð
```Ð
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### 3. WebSocket — signaling notifications ONLY (no audio)Ð
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Connect on login (keep alive):Ð
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```Ð
wss://api.shikkhasomoy.com/ws/announce?token={JWT}&role=controllerÐ
```Ð
Ð
**Listen for JSON:**Ð
Ð
| type | Action |Ð
|-----|-----|Ð

```

```

| `live_answer` | `{ session_id, answer: { sdp, type } }` !' setRemoteDescription |Ð
| `live_ice` | `{ session_id }` !' poll GET `/api/live/ice?role=controller` |Ð
| `live_end` | `{ session_id, reason }` !' teardown WebRTC, update UI |Ð
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**Do NOT send binary audio over WebSocket.**Ð
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#### 4. WebRTC Manager — LiveWebRtcController.ktÐ
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**On Start Live:**Ð
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1. `POST /api/live/session` !' get `session_id` + `ice_servers`Ð
2. Create `PeerConnectionFactory` with Opus, echo cancellation, noise suppression, AGC ONÐ
3. Create `PeerConnection` with ICE serversÐ
4. Create `AudioSource` + `LocalAudioTrack` from micÐ
5. Add local audio track to PeerConnectionÐ
6. `createOffer()` !' `POST /api/live/offer`Ð
7. On ICE candidate !' `POST /api/live/ice`Ð
8. On WS `live_answer` or poll GET `/api/live/session` !' `setRemoteDescription(answer)`Ð
9. Poll GET `/api/live/ice` for player candidates !' `addIceCandidate()`Ð
Ð
**On Stop Live:**Ð
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1. Stop local audio trackÐ
2. Close PeerConnectionÐ
3. `POST /api/live/end`Ð
4. Reset UIÐ
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**Mute:** `localAudioTrack.setEnabled(false/true)`Ð
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**Connection states:** Connecting / Live / Disconnected / FailedÐ
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**Auto reconnect:** on failure, end session and allow restart (max 3 retries with backoff)Ð
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#### 5. UI — Live Announcement ScreenÐ
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- **Start Live** !' runs WebRTC flowÐ
- **Stop Live** !' teardownÐ
- **Mute / Unmute**Ð
- Status: "Waiting for player" / "Connecting..." / "Live" / "Disconnected"Ð
- Error toast: 401, 403, 410, ICE failed, no player onlineÐ
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#### 6. Flow DiagramÐ
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User taps Start LiveÐ

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!' POST /api/live/sessionĐ
!' WebRTC createOfferĐ
!' POST /api/live/offerĐ
!' [Player creates answer]Đ
!' WS live_answer OR poll GET /api/live/sessionĐ
!' setRemoteDescription(answer)Đ
!' ICE exchange (POST + GET /api/live/ice)Đ
!' PeerConnection CONNECTED !' mic audio P2P to PlayerĐ
```

```
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User taps Stop LiveĐ
```

```
!' POST /api/live/endĐ
!' close PeerConnectionĐ
```

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## RulesĐ
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- ```
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- Match existing project code style, package names, Retrofit patternsĐ
- Reuse existing JWT token storageĐ
- Do not change unrelated featuresĐ
- Handle Android 12+ foreground service if mic runs in backgroundĐ
- Test on real device (emulator WebRTC may be limited)Đ
```

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```

```
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DeliverablesĐ
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- ```
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1. `LiveApiService.kt` (Retrofit interface)Đ
2. `LiveWebRtcController.kt` (WebRTC sender logic)Đ
3. Updated Live Announcement UI screenĐ
4. Updated WebSocket handler (controller = JSON only, no PCM)Đ
5. Remove all old PCM-over-WebSocket live codeĐ
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```

```
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## First Step for CursorĐ
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Search the codebase for existing live announcement / WebSocket / PCM / mic streaming code and show which files will change. Then implement step by step.Đ
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*School Bell Backend v2.0 — WebRTC Signaling API*Đ
```